

Nilay Jadav

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Embedded systems engineer specializing in FPGA-based platforms, real-time embedded software in C/C++, and hardware-software co-design for sensing, medical, and aerospace applications.

SKILLS

Software: C, C++, Embedded C, Python, MATLAB, Embedded Linux, Verilog, Assembly, Shell Scripting, Makefiles

Hardware: FPGA, Zynq SoC, ARM, AVR, PIC, Jetson boards, RPi, Power Electronics, Sensors (I2C, SPI, UART, RS232), PCB Design

Tools & Platforms: Vivado, Vivado HLS, Vitis, STM32 CubeIDE, Arduino IDE, KiCad, Atmel studio, Proteus, Keil, Altium Designer, Simulink, VS Code, Git, JIRA, Wireshark, Fusion 360, Debugging (JTAG, ILA, Oscilloscope, Logic Analyzer)

EDUCATION

Master of Engineering in Embedded Systems

May 2026

University of Maryland, College Park, MD

GPA: 3.95/4.0

Relevant Coursework: Embedded Systems Hardware, Embedded Systems Software, Networking and Distributed Systems 5G/6G, Real Time Operating Systems, Autonomous Robotics, IoT, Embedded System Hacking and Security

Bachelor of Technology in Electronics and Communication Engineering

April 2024

Charusat University, Anand, Gujarat, India

GPA: 9.20/10

EXPERIENCE

NeurQi (Weinberg Medical Physics) | North Bethesda, MD (Onsite)

Oct 2025 – May 2026

Electrical Engineer Intern

- Implemented Zynq multi-FPGA (50 boards) based control firmware for MRI coil actuation, timing synchronization, and safety monitoring, enabling precise and reliable brain imaging for early detection and treatment of dementia.
- Designed high-efficiency power supply modules for the Mobile MRI system (Ladybug), ensuring safety requirements.
- Developed application-level software, internal tools, and documentation to support system integration and improve readiness for human clinical testing, achieving nanosecond-level synchronization across control modules.

Mechanical Department, UMD (with Dolby Laboratories) | College Park, MD (Onsite)

August 2024 – January 2025

Graduate Research Assistant

- Conducted industry-sponsored research, building and analyzing neural network models using reservoir computing.
- Implemented a stereo audio algorithm in MATLAB achieving 45% data compression, contributing to development-focused signal processing and algorithm optimization efforts.

Space Applications Center (ISRO) | Ahmedabad, India (Onsite)

December 2023 – May 2024

Researcher

- Developed GNSS-based satellite positioning algorithms on a space-grade FPGA board using Verilog.
- Created custom hardware accelerators in HLS and integrated via AXIS and DMA protocols, enabling real-time data ingestion from MATLAB.
- Optimized matrix multiplication kernels on FPGA, achieving more than 50x improvement in throughput for real-time GNSS positioning data processing.

AethonLabs LLC | San Francisco, CA (Remote)

March 2023 – September 2023

IoT Intern

- Researched and developed proof-of-concept IoT products including a Smart HUD Helmet and a Smart Garage Door.
- Built an ESP32-based wireless keypad and control unit enabling multi-PIN access, scheduling, and smart-home integration.

Motorola Solutions | Texas, USA (Remote)

July 2022 – November 2022

Student Intern

- Developed C/C++ projects for streaming protocols and real-time embedded data processing, improving system reliability and latency handling in embedded workflows.

PROJECTS & PATENTS

Correlation Power Analysis (CPA) on STMF4

March 2025

- Conducted Python-based CPA attack on AES-128 hardware traces, recovering full keys with fewer than 5,000 traces.
- Extended to second-order masked AES, uncovering weaknesses and publishing reproducible analysis.

Self-driving Robot for Power Gathering Applications

April 2024

- Built custom AI image detection model on Jetson Orin and a power harvesting system using super-capacitors.
- Secured \$1,150 in funding and led team to 1st Prize (\$2,400) at MYRT International Robotics Competition.

IoT-Blockchain-based Water Quality & Monitoring

July 2024

- Built a blockchain-based water drone for decentralized secure monitoring using Jetson Orin and industrial sensors.
- Secured \$1,000 in funding from SSIP (Student Startup and Innovation Policy).

IoT based Smart Irrigation Nodes

July 2022

- Developed ESP32 irrigation nodes with web access, mobile application and LoRa based transmission.

PATENT: "SightSense" Smart Assistive Blind Stick

2024

- Designed and built a 3D printed ML based assistive navigation device with object detection, Bluetooth audio and vibration feedback.
- Secured \$2,400 in government funding.

Paper: FPGA implementation of Hardware Accelerated 64-bit Floating Point Matrix-Multiplier for GNSS Data Processing 2024

Paper: AI-Enhanced Quad-Wheeled Robot for Targeted Plant Disease Surveillance in Greenhouses 2023